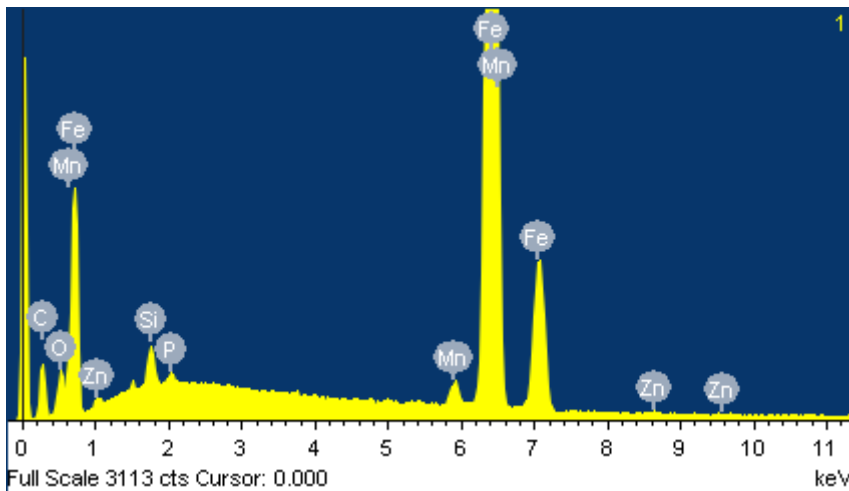
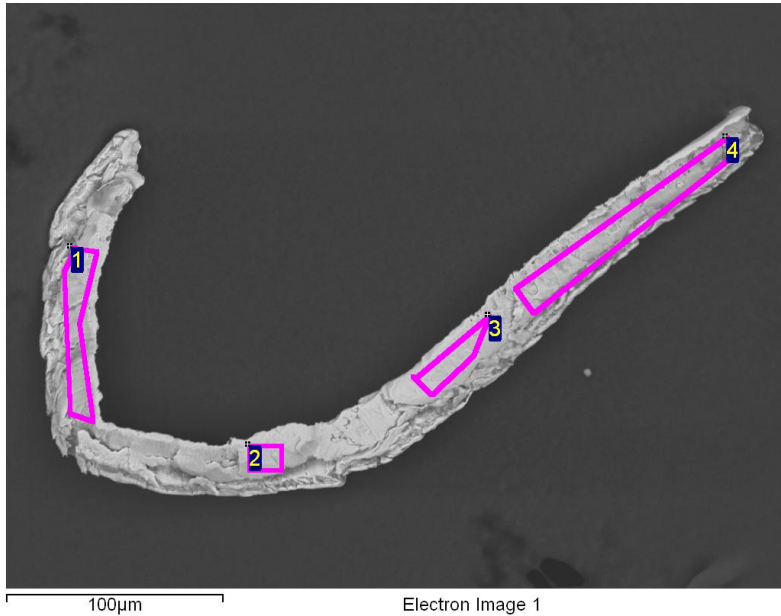


PS/QMM8-NaP		Material analysis report	
Report Date	: 21.07.2026	Report No.	: CC-CRI-QMM2-FI260716
Production		0 Km	
Investigation	✓	Field	✓
Issued By	: PS/QMM8-NaP		
Distribution:			
Subject	: Material grade and probable source identification of particles.		
1) Task	: Material analysis and dimensional measurement of 01 particles.		
2) Performed tests			
1. Scanning Electron Microscope	✓		
2. EDAX - Analysis	✓		
3. Optical Microscope	✗		
4. Hardness measurement	✗		
3) Sample preparation for SEM and EDAX:			
Particles placed on carbon tape stub.			
4) Sample preparation for microstructure investigation :			
5) Calibration			
EDAX Unit was calibrated with cobalt reference standard. EDAX analysis was performed under similar conditions with the following SEM/EDAX parameters.			
EHT: 20KV	WD: 8.5mm	Spot size :	
Signal: SE/BSD	Dead time :30-40 %	Live time : 50'	
6) Sample details			
Requisition Dept.	QMM2		
Requisition dept complaint No.	CRI.I. 26-130		
External Customer	M&M		
Injector no.	0445 111 113		
Mfd & Sr. no..	54005-1323		
Location of particle	NB seat Area		
Sample Submission date :	13-07-26		
Prepared by	Released by		
N R Khairnar	J R Nagrale PS/QMM8-NaP		

Project: Field CRI.I. 26-130 NB Seat Area Particle & Ball Damage (M&M)11
 Owner: MICO_BOSCH_NaP_QMM8
 Site: Site of Interest 1

Sample: Sample 1
 Type: Default
 ID:



Processing option : All elements analysed (Normalised)

Spectrum	In stats.	C	O	Si	P	Mn	Fe	Zn
1	Yes	8.92	2.58	0.90	0.28	1.41	85.45	0.4'
2	Yes	4.44	0.99	0.63		1.46	92.48	
3	Yes	3.81	0.99	0.52		1.52	93.16	
4	Yes	15.82	2.13	0.20		1.33	79.87	0.6:
Max.		15.82	2.58	0.90	0.28	1.52	93.16	0.6:
Min.		3.81	0.99	0.20	0.28	1.33	79.87	0.4'

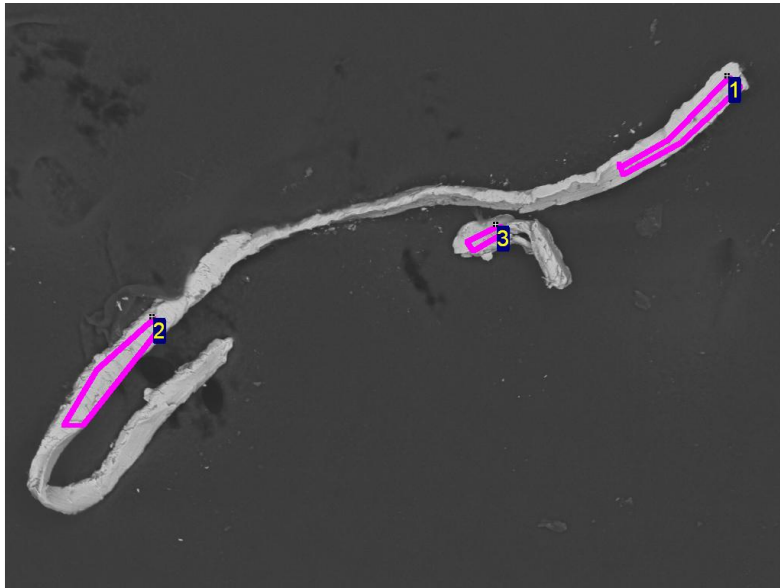
All results in weight%

Comment:

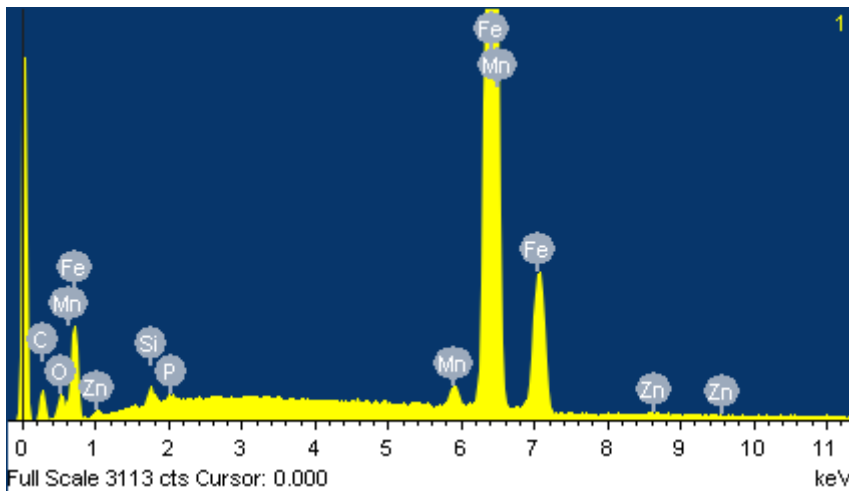
Particle Details																													
Length in μm	600																												
Width in μm	30																												
Chemistry	1.5 Mn steel																												
Surfacing coating	Nil																												
Probable Injector material source	Magnet Nut , NR nut??																												
Probable Indirect material source at NaP	★ ... BaP Line <table><tr><td>7</td><td>Ball Guide Plate</td><td></td><td>4.2 Station</td></tr><tr><td>24</td><td>Shim Position</td><td></td><td>5.3 Station</td></tr><tr><td>40</td><td>Co-Axiality Gauge</td><td></td><td>P1 Station</td></tr></table> <table><tr><td>24</td><td>Shim Position</td><td></td><td>5.3 Station</td></tr><tr><td>40</td><td>Co-Axiality Gauge</td><td></td><td>P1 Station</td></tr></table> Cleaning area <table><tr><td>17</td><td>Magnet Nut Spanner</td><td></td><td>Dis-Assembly Bench</td></tr></table> Cleaning area <table><tr><td>24</td><td>Valve Set removing Base</td><td></td><td>Dis-Assembly Bench</td></tr></table>	7	Ball Guide Plate		4.2 Station	24	Shim Position		5.3 Station	40	Co-Axiality Gauge		P1 Station	24	Shim Position		5.3 Station	40	Co-Axiality Gauge		P1 Station	17	Magnet Nut Spanner		Dis-Assembly Bench	24	Valve Set removing Base		Dis-Assembly Bench
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24	Valve Set removing Base		Dis-Assembly Bench																										
Probable source from pump																													
Probable external material source	HPP,																												
Remarks																													

Project: Field CRI.I. 26-130 NB Seat Area Particle & Ball Damage (M&M)11
 Owner: MICO_BOSCH_NaP_QMM8
 Site: Site of Interest 2

Sample: Sample 1
 Type: Default
 ID:



Electron Image 1



Processing option : All elements analysed (Normalised)

Spectrum	C	O	Si	P	Mn	Fe	Zn	Total
1	5.39	1.46	0.43	0.22	1.46	90.69	0.35	100.0
2	5.98	1.05	0.37	0.17	1.27	90.88	0.27	100.0
3	11.00	3.23	0.92	0.35	1.25	82.52	0.73	100.0
Mean	7.46	1.91	0.57	0.25	1.33	88.03	0.45	100.0
Std. deviation	3.09	1.16	0.30	0.09	0.11	4.78	0.24	
Max.	11.00	3.23	0.92	0.35	1.46	90.88	0.73	
Min.	5.39	1.05	0.37	0.17	1.25	82.52	0.27	

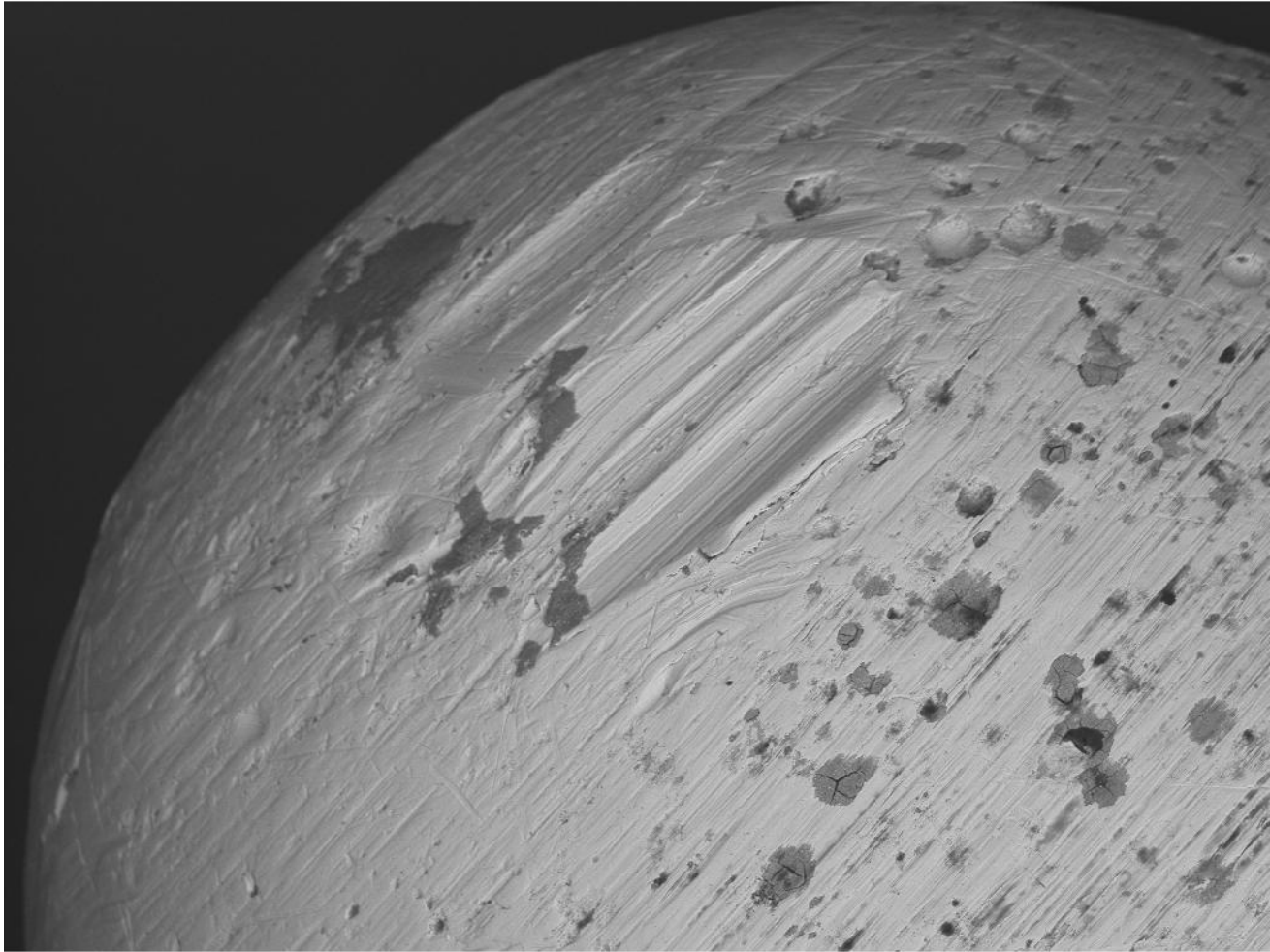
All results in weight%

Comment:

Particle Details																													
Length in μm	1050																												
Width in μm	25																												
Chemistry	1.5 Mn steel																												
Surfacing coating	Nil																												
Probable Injector material source	Magnet Nut , NR nut??																												
Probable Indirect material source at NaP	★ . . . BaP Line <table><tr><td>7</td><td>Ball Guide Plate</td><td></td><td>4.2 Station</td></tr><tr><td>24</td><td>Shim Position</td><td></td><td>5.3 Station</td></tr><tr><td>40</td><td>Co-Axiality Gauge</td><td></td><td>P1 Station</td></tr></table><table><tr><td>24</td><td>Shim Position</td><td></td><td>5.3 Station</td></tr><tr><td>40</td><td>Co-Axiality Gauge</td><td></td><td>P1 Station</td></tr></table> Cleaning area <table><tr><td>17</td><td>Magnet Nut Spanner</td><td></td><td>Dis-Assembly Bench</td></tr></table> Cleaning area <table><tr><td>24</td><td>Valve Set removing Base</td><td></td><td>Dis-Assembly Bench</td></tr></table>	7	Ball Guide Plate		4.2 Station	24	Shim Position		5.3 Station	40	Co-Axiality Gauge		P1 Station	24	Shim Position		5.3 Station	40	Co-Axiality Gauge		P1 Station	17	Magnet Nut Spanner		Dis-Assembly Bench	24	Valve Set removing Base		Dis-Assembly Bench
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Probable external material source	HPP,																												
Remarks																													

Project: Field CRI.I. 26-130 NB Seat Area Particle &
Ball Damage (M&M)11
Owner: MICO_BOSCH_NaP_QMM8
Size: Size of Image 2 Ball Damage

Sample: Sample 1
Type: Default
ID:



400µm

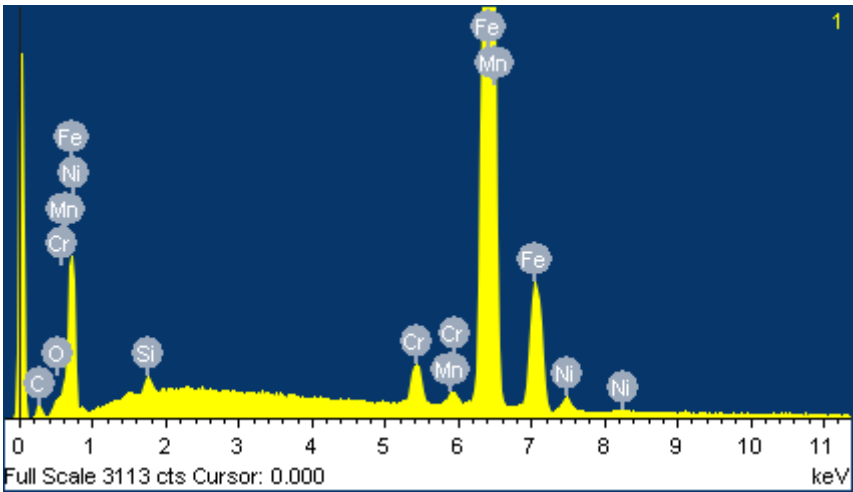
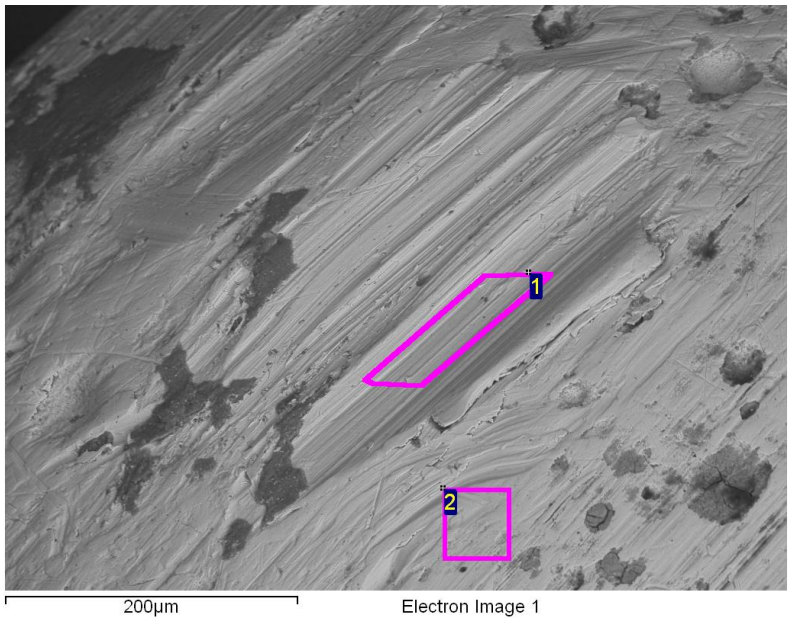
Electron Image 1

Comment:

Project: Field CRI.I. 26-130 NB Seat Area Particle & Ball Damage (M&M)11
Owner: MICO_BOSCH_NaP_QMM8
Site: Site of Interest 4 Ball Damage

Sample: Sample 1
Type: Default
ID:

Comment:
External damage on ball tip



Processing option : All elements analysed (Normalised)

Spectrum	In stats.	C	O	Si	Cr	Mn	Fe	Ni	Total
1	Yes	2.42	1.09	0.48	2.01	0.69	91.40	1.90	100.00
2	Yes	3.23	1.80	0.38	2.27	0.64	89.69	1.99	100.00
Mean		2.83	1.44	0.43	2.14	0.67	90.54	1.95	100.00
Std. deviation		0.57	0.50	0.07	0.19	0.04	1.21	0.06	
Max.		3.23	1.80	0.48	2.27	0.69	91.40	1.99	
Min.		2.42	1.09	0.38	2.01	0.64	89.69	1.90	

All results in weight%